- In p-block elements, the first member of each group differs from the other elements of the corresponding group.
- In heavier post-transition metals, the outer s electrons (ns) have a tendency to remain inert and show reluctance to take part in the bonding, which is known as inert pair effect.
- Some elements exist in more than one crystalline or molecular forms in the same physical state. For example, carbon exists as diamond and graphite. This phenomenon is called allotropism
- Borax is a sodium salt of tetraboric acid. It is obtained from colemanite ore by boiling its solution with sodium carbonate.
- Boric acid can be extracted from borax and colemanite.
- Boric acid has a two dimensional layered structure.
- The name alum is given to the double salt of potassium aluminium sulphate [K_2SO_4. $Al_2(SO_4)_3$.24.H_2O].
- Carbon is found in the native form as graphite.
- Silicon occurs as silica (sand and quartz crystal). Silicate minerals and clay are other important sources for silicon.
- Catenation is the ability of an element to form chain of atoms
- Carbon nanotubes, another recently discovered allotropes, have graphite like tubes with fullerene ends.
- Silicones or poly siloxanes are organo silicon polymers with general empirical formula (R_2SiO). Because of their very high thermal stability they are called high –temperature polymers.
- The mineral which contains silicon and oxygen in tetrahedral $[SiO_4]^{4-}$ units linked together in different patterns are called silicates.
- Types of Silicates:
  - **Ortho silicates (Neso silicates), Pyro silicate (or) Soro silicates),** Cyclic silicates (or Ring silicates)
  - Inosilicates : **Chain silicates (or pyroxenes), Double chain silicates (or amphiboles):**
  - Sheet or phyllo silicates
  - Three dimensional silicates (or tecto silicates)
- Zeolites are three-dimensional crystalline solids containing aluminium, silicon, and oxygen in their regular three dimensional framework.
- Zeolites act as a molecular sieve for the removal of permanent hardness of water

XII_U2-P-Block.indd 52 22-12-2022 20:06:01